Implications of dynamic mass on galactic rotation curves

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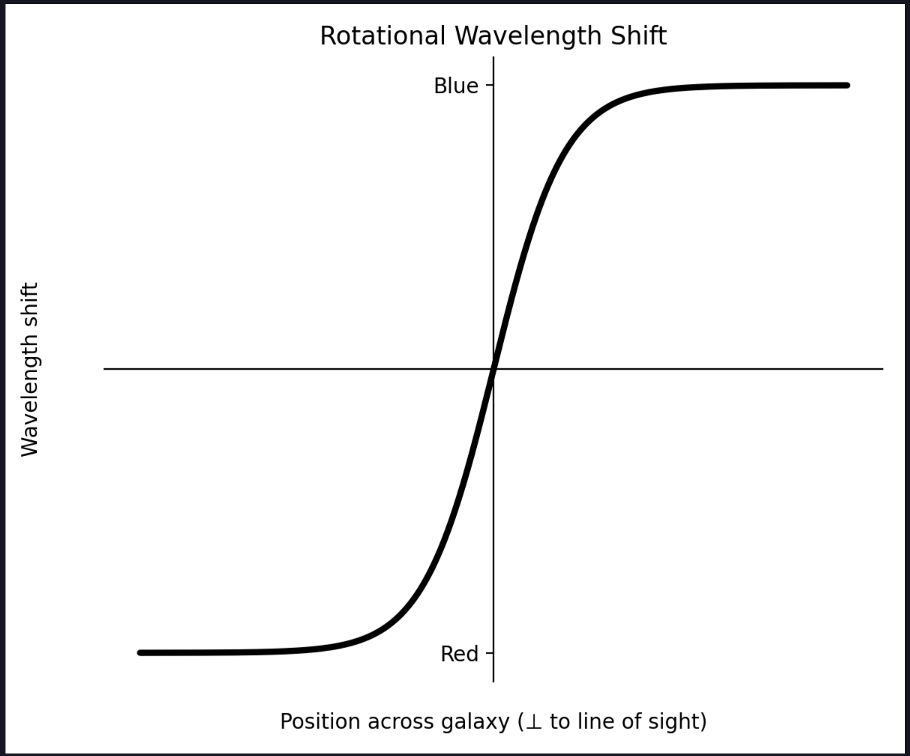
Overview

- What I'm working on
- Background GR
- Where I'm at

Galactic rotation curves --> rotational velocity gradient

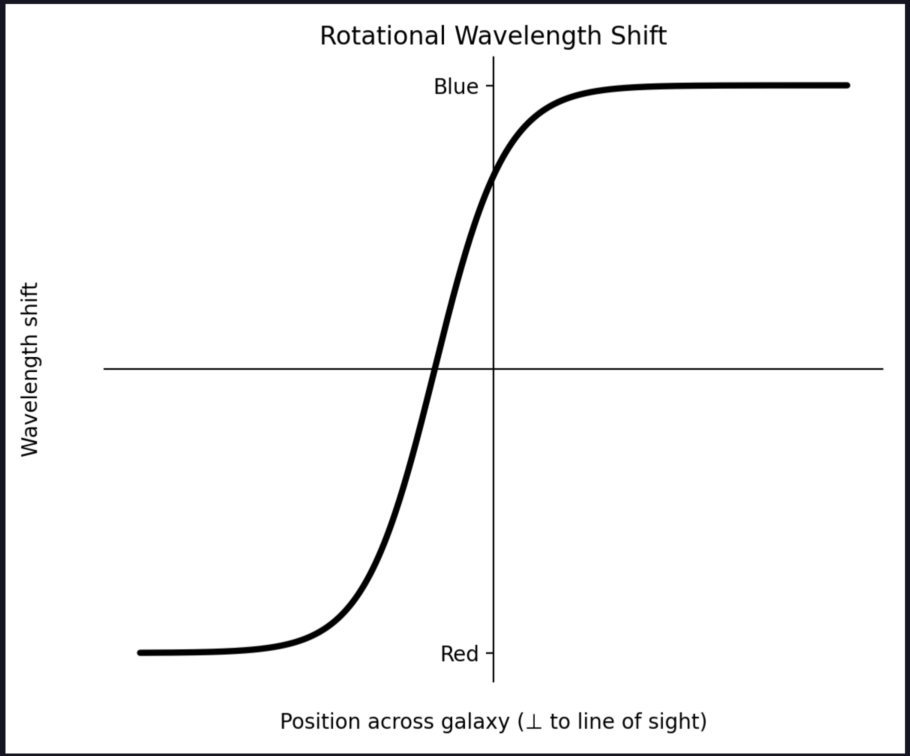
- rotating galaxies produce a position-dependent wavelength (Doppler) shift due to rotation



Right side is rotating towards the observer (redshifted)

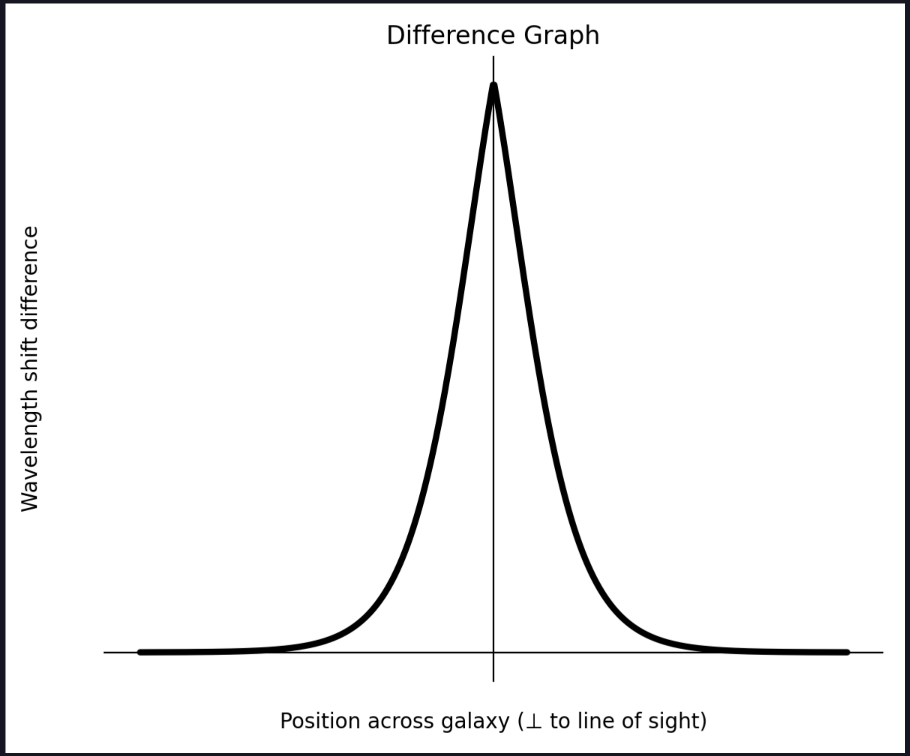
Spectroscopy issue

- In more than 50% of spiral galaxies, the kinematic and luminous centers do not overlap [3].



difference in luminosity curves

The shift is indicative of a function of the form



solution possibility

It was shown in [2] that a non-constant mass leads to universal acceleration changes as well as shifts in spectra.

GR background

- Space time interval

$$
\begin{aligned}
ds^2 =& dt^2 - dx^2 - dy^2 - dz^2 \\
=& g_{\mu\nu} dx^\mu dx^\nu
\end{aligned}
$$

Einstein summation convention

- flat space

$$
g_{\mu\nu} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}
$$

GR background cont.

- timelike $ds^2 > 0 \Rightarrow ds^2 = d\tau^2$
- $S = \int d\tau = \int \sqrt{ds^2} = \int \sqrt{\frac{ds^2}{d\lambda^2}} d\lambda = \int \sqrt{g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}} d\lambda = \int \mathcal{L} d\lambda$

Schwarzschild

Schwarzschild as estimate of low/non-rotating body in space:

$$g_{\mu\nu} = \begin{bmatrix} \left(1 - \frac{r_s}{r}\right) & 0 & 0 & 0 \\ 0 & -\left(1 - \frac{r_s}{r}\right)^{-1} & 0 & 0 \\ 0 & 0 & -r^2 & 0 \\ 0 & 0 & 0 & -r^2 sin^2(\theta) \end{bmatrix}$$

Schwarzschild Cont.

$$S = m \int \sqrt{g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}} d\lambda = \int \mathcal{L} d\lambda$$

$$\mathcal{L} = m \sqrt{(1 - \frac{r_s}{r})(t')^2 - (1 - \frac{r_s}{r})^{-1}(r')^2 - r^2(\phi')^2}$$

prime indicates a partial derivitive w.r.t. the affine parameter λ

dynamic mass Schwarzschild

$$\mathcal{L} = m(r)\sqrt{(1-\frac{r_s}{r})(t')^2 - (1-\frac{r_s}{r})^{-1}(r')^2 - r^2(\phi')^2}$$

prime indicates a partial derivitive w.r.t. the affine parameter λ

variable mass Schwarzschild

Equation of motion:

$$\frac{\partial m}{\partial r}\left(1-\frac{r_s}{r}-\dot{r}^{\,2}\right)=m(r)\left(\ddot{r}+\frac{r_s}{2\left(r-r_s\right)r}\dot{r}^2+(r-r_s)\dot{\phi}^2-\frac{\left(1-\frac{r_s}{r}\right)r_s}{2r^2}\dot{t}^2\right)$$

$$\frac{\partial m}{\partial r}\left(1-\frac{r_s}{r}-\dot{r}^{\,2}\right)=m(r)\left(\ddot{r}+\frac{r_s}{2\left(r-r_s\right)r}\dot{r}^2\right)+\frac{\left(1-\frac{r_s}{r}\right)}{m(r)}\left(\frac{L^2}{r^3}-\frac{r_sE^2}{2r^2}\right)$$

dot indicates derivive with respect to proper time

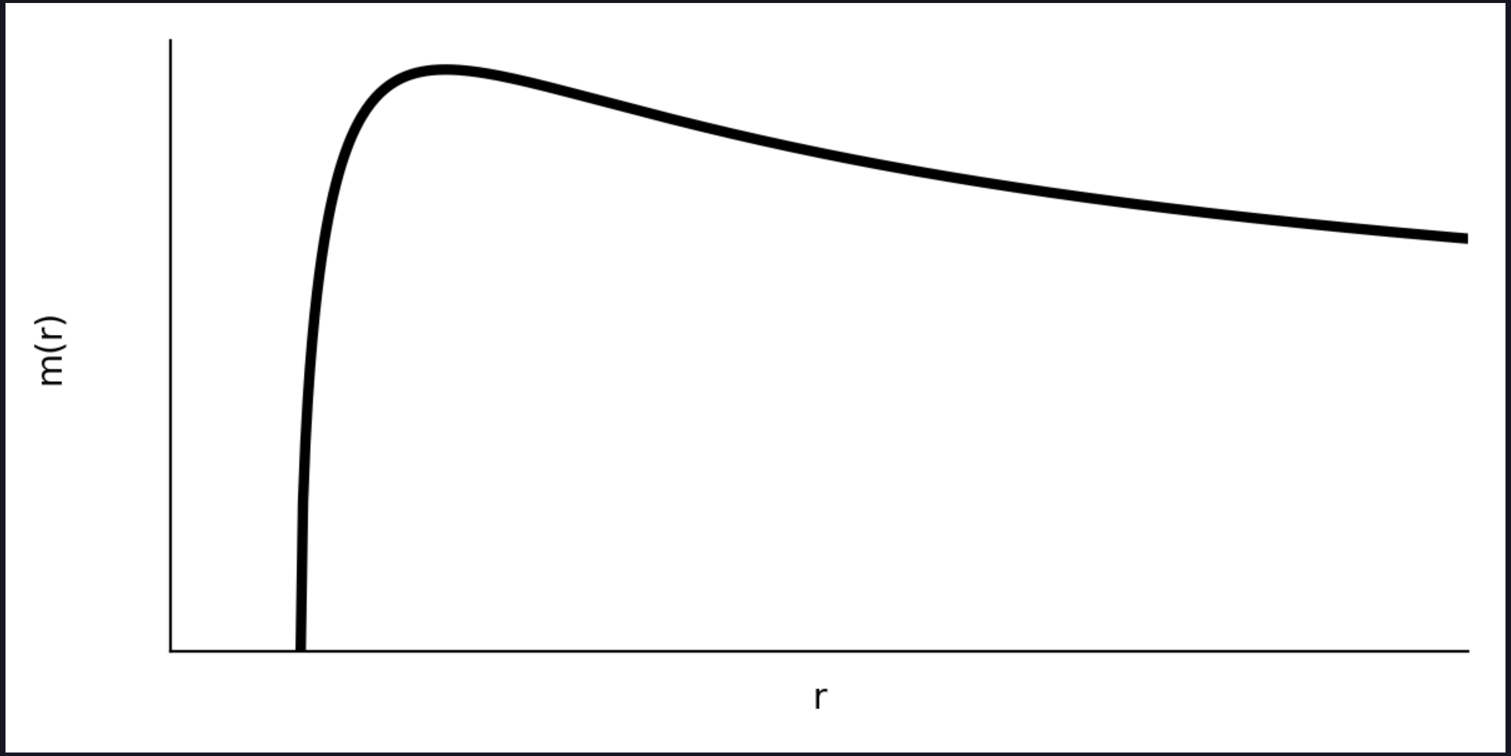
cicular orbit

$$\dot{r} = 0, \quad \ddot{r} = 0$$

$$\frac{\partial m}{\partial r} = \frac{1}{m(r)} \left(\frac{L^2}{r^3} - \frac{r_s E^2}{r^2} \right)$$

$$m(r) = \sqrt{c + \frac{2E^2}{r} - \frac{L^2}{r^2}}$$

cicular orbit cont.



references

[1] Jog, Chanda J., and Francoise Combes. "Lopsided Spiral Galaxies." Physics Reports 471, no. 2 (2009): 75–111. https://doi.org/10.1016/j.physrep.2008.12.002.

[2] Shi, Yuan. "Force, Metric, or Mass: Disambiguating Causes of Uniform Gravity." arXiv:1908.02159. Preprint, arXiv, September 3, 2021. https://doi.org/10.48550/arXiv.1908.02159.

[3] van Eymeren, J., Jütte, E., Jog, C. J., Stein, Y., and Dettmar, R.-J. "Lopsidedness in WHISP Galaxies. I. Rotation Curves and Kinematic Lopsidedness." Astronomy & Astrophysics 530 (2011): A29. https://doi.org/10.1051/0004-6361/201016177.